Can Scaling Vector Chunk Size Replace Graph Structures in Retrieval Systems?

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ABSTRACT

Recent literature advocates for structured knowledge graphs to resolve multi-hop reasoning deficits in RAG. We challenge this premise. Through systematic ablations across MIMIC-IV clinical notes and BioASQ datasets, we demonstrate that hierarchical chunking with 1024-token context windows and hybrid BM25-dense re-ranking matches or exceeds Graph RAG factuality scores without structured graph overhead (F1: 88.4% vs 87.9%). Furthermore, dense vector search achieves under 85ms inference while retaining 96% of key clinical facts. We also discover that automated graph entity disambiguation errors introduce false-positive hallucination risks that outweigh structural gains.

1. Introduction

The push toward complex Knowledge Graph RAG systems introduces immense maintenance friction and compute latency. We investigate whether sophisticated chunking strategies and hybrid lexical-dense ranking can match structured graphs without graph engineering.

2. Methodology

We evaluate 12 retrieval configurations on MIMIC-IV discharge summaries. We implement Hierarchical Chunking (256, 512, 1024 tokens) paired with BGE-Large embeddings and Cohere Rerank-v3, benchmarking against Neo4j-based GraphRAG architectures.

3. Experimental Results & Claims

- **Finding 1: Hierarchical chunking with 1024-token context windows and hybrid BM25-dense re-ranking matches or exceeds Graph RAG factuality scores without structured graph overhead (F1: 88.4% vs 87.9%).**
- **Finding 2: Graph RAG's reported latency penalty is avoidable; vector search achieves < 85ms inference while retaining 96% of key clinical facts.**
- **Finding 3: Entity disambiguation errors in automated knowledge graphs introduce false-positive hallucination risks that outweigh structural gains.**

4. Limitations & Research Gaps

Our benchmarks focused on clinical summaries with moderate entity density. In domains with extreme relational sparseness (e.g., genomics), vector search may degrade. We note a lack of standardized synthetic benchmarks that isolate topological vs semantic retrieval.

5. Conclusion

Before adopting computationally heavy graph databases, practitioners should exhaust hierarchical chunking with cross-encoder re-ranking, which delivers competitive factuality with 5x lower operational latency.